Sprint 3: Market Data and Conversion Routes

How SLE turns provider facts into auditable reference rates

In progress

Starting status: the module specification and pricing architecture decision are established. Database schema, adapters, evaluation logic, worker refresh, safety guards, and test evidence are not implemented yet.

What Came Before

Sprint 1 taught SLE which assets, fiat currencies, networks, and markets exist. Sprint 2 made private requests and background events authenticated, retry-safe, and durable. Sprint 3 can now collect market facts without hardcoding ETH, USDT, USD, or NGN.

Goal in Simple Terms

A provider may tell SLE that one ETH is worth a number of USDT. Another source may describe USDT in USD, and a reviewed manual version may describe USD in NGN for the free sandbox. Sprint 3 stores each fact and follows a configured recipe to calculate a reference rate such as ETH/NGN.

The Three Main Records

Record	Junior-friendly meaning
Observation	One immutable provider fact with its price and time.
Conversion route	An ordered recipe saying which pairs to multiply or divide.
Reference snapshot	The calculated result plus links to every observation used.

Example Flow

```
Coinbase: ETH/USDT observation
          | multiply
Stablecoin guard: USDT/USD observation
          | multiply
Manual version: USD/NGN observation
          v
Configured ETH/NGN reference snapshot
          |
          v
Sprint 4 quote engine adds spread and purchase fee
```

> **Important boundary:** Sprint 3 does not create a customer quote. It produces a reference rate. Sprint 4 owns the customer rate, spread, fees, limits, and expiry.

Exact Arithmetic

Provider prices enter as decimal strings. Route math uses `decimal.js`, never JavaScript floating point. Intermediate legs retain working precision. The final rate is normalized once to a configured scale using recorded half-even rounding. Customer principal is not rounded here.

Safety Checks

- **Staleness:** old or missing observations cannot price a route.
- **Deviation:** an unexpectedly large move from the last accepted snapshot is rejected.
- **Sequence gap:** missing or backwards provider sequences make the pair unsafe.
- **Stablecoin guard:** a configured stablecoin must remain inside its explicit reference-fiat band.

SLE never assumes an asset is stable merely because its symbol is USDT or USDC.

System Link

Provider adapters remain behind a domain interface. PostgreSQL stores evidence and route versions. The worker refreshes observations and snapshots. Sprint 4 receives an accepted snapshot ID, while later reconciliation can reproduce exactly which source facts supported a quote.

Free MVP Limitation

The sandbox may use Coinbase public reference data and a reviewed manual fiat cross-rate. This demonstrates the flow but is not approved for real funds. Independent redundant production sources remain a launch gate.

Planned Evidence

- Exact direct and multi-leg arithmetic tests.
- Stale, missing, deviation, sequence-gap, and depeg rejection tests.
- Real PostgreSQL immutability, uniqueness, atomic snapshot, and concurrency tests.
- Provider timeout and worker restart evidence.
- A final updated Sprint 3 report and regenerated PDF.

Glossary

Pair: two instruments whose relative price is observed. **Leg:** one step in a conversion route. **Cross-rate:** a rate calculated through intermediate pairs. **Depeg:** a stablecoin moving outside its expected reference band. **Basis point:** one hundredth of one percent.